Texas A&M University
College of Liberal Arts
Department of Economics
Intermediate Macroeconomics, ECON 410
Spring 2026
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Midterm 2

March 24th, 2026

On my honor, as an Aggie, I have neither given nor received unauthorized aid on this academic work

Signed: ______________________________

NAME: ______________________________

UIN: ______________________________

30 questions in total. All questions have equal weight.
Double check your answers. Submit the gradescope only.

The money demand equation for the Quantity Theory of Money is:

$$\frac{M_t}{P_t} = \frac{Y_t}{v_t}$$

Suppose that the money demand equation for the Cagan model is:

$$\frac{M_t}{P_t} = \frac{Y_t}{(1 + i_t)}.$$

1. According to the Quantity Theory of Money, a permanent increase in money velocity from v_0 to v_1 will increase inflation permanently.

   (a) True

   (b) False

2. According to the Solow growth model, poorer countries grow faster than richer countries if they have the same saving rate, depreciation rate, population growth, and technology.

   (a) True

   (b) False

3. In the Solow growth model, we denote per worker variables by:

   (a) Capital letters

   (b) small letters

4. If nominal wages cannot be cut, then the only way to reduce real wages is by:

   (a) adjustments via inflation.

   (b) unions.

   (c) legislation.

   (d) productivity increases.

5. The steady-state level of capital occurs when the change in the capital stock per worker (Δk) equals:

   (a) 0.

   (b) the saving rate.

   (c) the depreciation rate.

   (d) the population growth rate.

6. Assume that two economies are identical in every way except that one has a higher saving rate. According to the Solow growth model, in the steady state the country with the higher saving rate will have ________ level of output per person and ________ rate of growth of output per worker compared to the country with the lower saving rate.

(a) the same; the same
(b) the same; a higher
(c) a higher; the same
(d) a higher; a higher

7. In the Cagan model, if the Fed announces that it will lower the money supply in the future but does not change the money supply today,

   (a) both the nominal interest rate and the current price level will decrease.
   (b) the nominal interest rate will increase and the current price level will decrease.
   (c) the nominal interest rate will decrease and the current price level will increase.
   (d) both the nominal interest rate and the current price level will increase.

8. Thao finds there are two assets available from her bank:

   (a) Asset A offers 1.05 oranges next year if you save 1 orange today, and
   (b) Asset B offers 4% annual nominal interest rate.

   After taking the ECON 410 class, Thao decides to invest in asset B. What *could be* her annual inflation expectations?

   (a) -1.5
   (b) -0.5
   (c) 0.5
   (d) 1.5

9. An economy with constant velocity has real GDP growth of 2 percent, money growth of 7 percent, and a real interest rate of 3 percent. The nominal interest rate is

   (a) 2%
   (b) 6%
   (c) 8%
   (d) 12%

10. An increase in the money supply by 500 percent must sooner or later bring about an increase in the price level of about the same proportion.

    (a) True
    (b) False

Based on this information, answer the following three questions. Consider an economy in which the demand for money is of the form

$$M_t = \frac{1}{v} P_t Y$$

for $t = 0, 1, 2, \ldots$, where output is 500, the money velocity is 2.5. The money supply was growing 1% for $t = 0, 1, 2$. At the start of period 3, the central bank unexpectedly announces that the money supply will grow at 3% from period 3 onward. In other words,

$$M_1 = (1.01)M_0, \quad M_2 = (1.01)M_1, \quad M_3 = (1.03)M_2, \quad M_4 = (1.03)M_3, \text{ and so on.}$$

11. The inflation rate in period 1 (π_1) is __________ and the real money balance in period 1 $\left(\frac{M_1}{P_1}\right)$ is __________.

    (a) 1.01%, 100
    (b) 1.01%, 200
    (c) 1%, 100
    (d) 1%, 200
    (e) 0%, 100

12. The expected inflation in period 2, given the information available in period 1 ($E_1\pi_2$) is __________ and the inflation rate in period 2 (π_2) is __________.

    (a) 0%, 1.01%
    (b) 1%, 3%
    (c) 1.01%, 1.01%
    (d) 1%, 1%
    (e) 1%, 1.01%

13. The expected inflation in period 3, given the information available in period 2 ($E_2\pi_3$) is __________ and the inflation rate in period 3 (π_3) is __________.

    (a) 0%, 1.01%
    (b) 1%, 3%
    (c) 1.01%, 1.02%
    (d) 1.01%, 3%
    (e) 1%, 2%

14. According to the Quantity Theory of Money, an unanticipated money-based inflation stabilization program that permanently reduces the money growth rate from 5 percent to 0 percent in the future causes deflation in the period the program is announced.

(a) True

(b) False

15. In the long run, according to the quantity theory of money, if velocity is constant, then __________ determines real GDP and __________ determines nominal GDP.

    (a) the productive capability of the economy; the money supply

    (b) the money supply; the productive capability of the economy

    (c) velocity; the money supply

    (d) the money supply; velocity

16. If there are 100 transactions in a year and the average value of each transaction is \$10, then if there is \$200 of money in the economy, transactions velocity is __________ times per year.

    (a) 0.2

    (b) 2

    (c) 5

    (d) 10

    (e) 20

Answer the following three questions based on the economy described by the below 5 equations. Consider Solow Growth model with constant population and technology, as follows.

$$Y_t = K_t^{0.5} L_t^{0.5}$$

$$Y_t = C_t + S_t$$

$$S_t = sY_t$$

$$K_{t+1} = (1 - \delta)K_t + I_t$$

$$S_t = I_t$$

where Y_t is output in period t, K_t is capital stock in period t, L_t is labor in period t, C_t is consumption in period t, S_t is savings in period t, and I_t is investment in period t. In this economy, a savings rate is 40 percent, and a depreciation rate is 10 percent.

17. Suppose that this economy starts with a capital stock per worker of 4 in period 1. The income per worker in period 1 is __________ and consumption per worker in period 1 is __________.

    (a) 1, 0.6

    (b) 2, 1.2

(c) 1, 0.7

(d) 2, 0.8

18. Suppose that this economy starts with a capital stock per worker of 4 in period 1. The capital stock per worker in period 2 is ________.

(a) 4.4

(b) 1.1

(c) 4.8

(d) 3.6

(e) 5

19. The steady-state stock of capital per worker is ________ and the steady-state income per worker is ________.

(a) 0.35, 0.7

(b) 1.84, 1.23

(c) 4, 2

(d) 16, 4

(e) 9, 3

Answer the following two questions using the table, which provides information on GDP per capita and population in the United States and China in 1980 and 2018. GDP per capita is measured in 2010 US dollars.

GDP per capita		
Year	U.S.	China
1980	28,589.7	347.1
2018	54,579.0	7752.6

20. The average annual growth rate of per capita GDP in the US between 1980 and 2018 is ________ %. The average annual growth rate of per capita GDP in China between 1980 and 2018 is ________ %. Round your solutions to the first decimal points, for example, 0.17 to 0.2.

(a) 1.8, 8.2

(b) 1.7, 8.8

(c) 1.7, 8.5

(d) 1.5, 8

(e) 1.8, 8.8

21. Now suppose that we were at the end of 2018. Based on the average annual growth rates of per capita GDP between 1980 and 2018 from above, what year would you expect China to catch up with the U.S.?

    (a) 2047
    (b) 2048
    (c) 2049
    (d) 2051

22. In the Solow growth model, if two countries are otherwise identical (with the same production function, same saving rate, same depreciation rate, and same rate of population growth) except that Country Large has a population of 1 billion workers and Country Small has a population of 10 million workers, the steady-state level of output per worker will be ____________ and the steady-state growth rate of output per worker will be ____________.

    (a) the same in both countries; the same in both countries
    (b) higher in Country Large; higher in Country Large
    (c) higher in Country Small; higher in Country Small
    (d) higher in Country Large; higher in Country Small

23. All debit and credit card transactions are included in the money supply.

    (a) True
    (b) False

24. Hyperinflations ultimately are the result of excessive growth rates of the money supply; the underlying motive for the excessive money growth rates is frequently a government's:

    (a) desire to increase prices throughout the economy.
    (b) need to generate revenue to pay for spending.
    (c) responsibility to increase nominal interest rates by increasing expected inflation.
    (d) inability to buy government securities through open-market operations.

25. According to the QTM, when the money supply decreases, __________ will decrease.

    (a) real GDP
    (b) consumption spending
    (c) the price level
    (d) investment spending

26. Consider a standard Solow growth model with production function

$$y = \sqrt{k}$$

and constant depreciation of 10% per year. The economy is currently in a steady state with saving rate $\sigma = 0.6$. It is known that the Golden Rule saving rate in this economy is $\sigma_{\text{gold}} = 0.5$. The government is considering reducing the saving rate from 0.6 to 0.5.

Which of the following statements is correct?

Hint: there is no need to derive c and y in any of the steady states.

(A) Both output per worker and consumption per worker will decrease in the long run.

(B) Output per worker will decrease in the long run, but consumption per worker will increase in the long run.

(C) Output per worker will increase in the long run, but consumption per worker will decrease in the long run.

(D) Both output per worker and consumption per worker will increase in the long run.

27. Because most loans are specified in nominal terms, low __________ inflation hurts __________.

(a) expected, debtors

(b) expected, creditors

(c) unexpected, debtors

(d) unexpected, creditors

28. In the Solow diagram below, indicate the possible location of the "golden rule" amount of capital

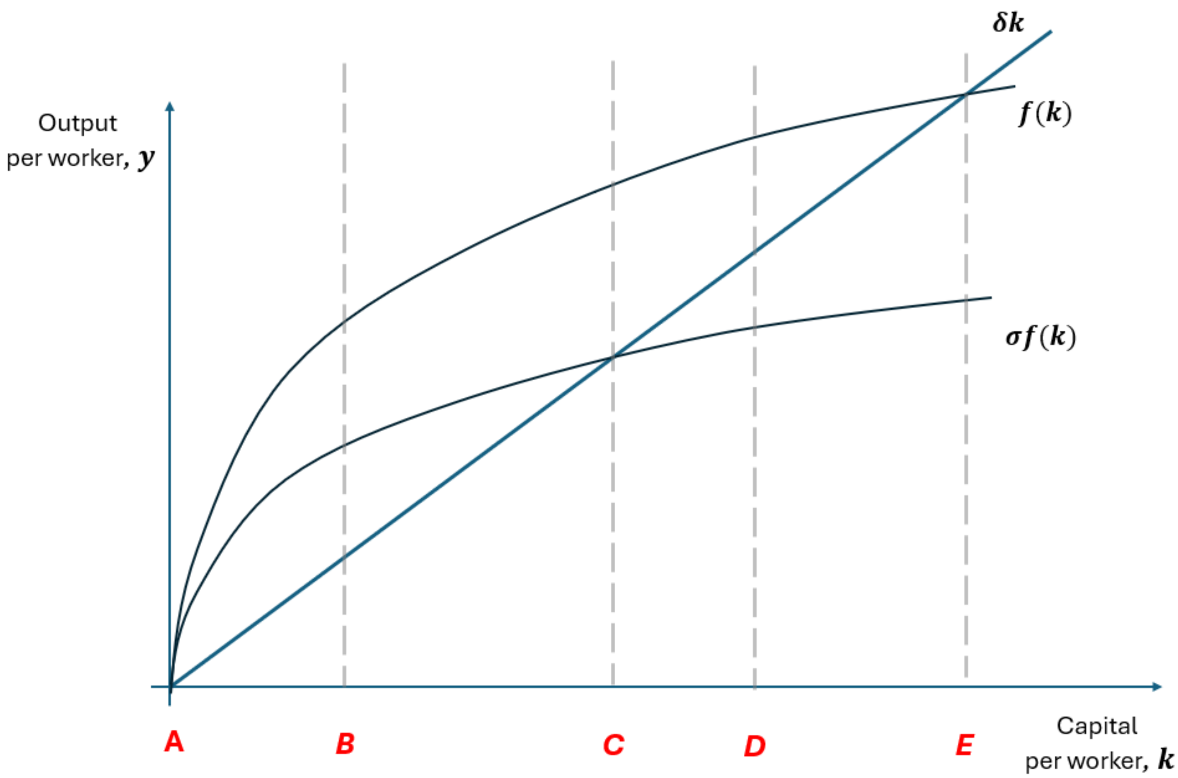
(a) point A

(b) point B

(c) point C

(d) point D

(e) point E



29. Suppose country A has the production technology $Y = \sqrt{KL}$, while country B has a better one: $Y = 2\sqrt{KL}$.

    Country B will have double the growth rate of output per worker in steady state in the basic Solow model compared to the growth rate of output per worker in steady state for country A.

    (a) True. For any given K and L, country B will produce twice as much output, thus in equilibrium the growth rate of economy B will also be twice as high as the growth rate of economy A.

    (b) False. In steady state, capital is not changing; therefore output is not changing, and the growth rate of output is zero in both cases, even though one economy is twice the size of the other.

30. In the basic Solow model, a war that destroys half of the capital stock, holding everything else constant, would lead to a higher growth rate of output per worker. *Assume that the economy was in the steady state before the war.*

    (a) True.

    (b) False.

You can use this page for your calculations: